

DEPARTMENT OF MATHEMATICS AND APPLIED MATHEMATICS

Mathematics II

Advanced Calculus (2AC)

Tutorial Sheet 12: Solutions

11th and 12th May 2023

1. We introduced vector fields $\vec{F} : D \subset \mathbb{R}^2 \rightarrow \mathbb{R}^2$ (called two-dimensional vector) and written $\vec{F} = (P, Q)$ or $\vec{F} : D \subset \mathbb{R}^3 \rightarrow \mathbb{R}^3$ (called three-dimensional vector field) written as $\vec{F} = (P, Q, R)$, where P, Q, R are the component functions of the vector field. We saw few examples in the lectures, among which the notion of gradient vector field also called conservative vector fields, that is,

$$\begin{cases} \vec{F}(x, y) = \nabla f(x, y) & \text{for some function } f : D \subset \mathbb{R}^2 \rightarrow \mathbb{R}, \\ \vec{F}(x, y, z) = \nabla f(x, y, z) & \text{for some function } f : D \subset \mathbb{R}^3 \rightarrow \mathbb{R} \end{cases}$$

That is, we seek $f : D \subset \mathbb{R}^2 \rightarrow \mathbb{R}$ such that $\begin{cases} f_x(x, y) = P(x, y) \\ f_y(x, y) = Q(x, y) \end{cases} \quad (*)$

- Integrate the first equation w.r.t. x and get

$$f(x, y) = \int P(x, y) dx + C(y). \quad (**)$$

- Differentiate $(**)$ w.r.t. y and get $f_y(x, y) = \frac{d}{dy} \left[\int P(x, y) dx \right] + C'(y)$.

- Use the second equation in $(*)$ to get that $C'(y) = Q(x, y) - \frac{d}{dy} \left[\int P(x, y) dy \right]$. - Find the function C and substitute it into $(**)$ to find the expression of the function f .

Similarly for 3D-vector fields!

Follow the steps above to show that each vector field below is conservative by finding a function f for which $\vec{F} = \nabla f$.

- (i) $\vec{F}(x, y) = (3x^2 + 6y, 3y^2 + 6x)$; We seek a differentiable function $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ such that $\nabla f(x, y) = \vec{F}(x, y)$ that is,

$$\begin{cases} f_x(x, y) = 3x^2 + 6y, \\ f_y(x, y) = 3y^2 + 6x. \end{cases}$$

Integrating the first equation w.r.t. x , we get

$$f(x, y) = \int (3x^2 + 6y) dx = x^3 + 6xy + C(y).$$

Now differentiating w.r.t. y , we get

$$3y^2 + 6x = f_y(x, y) = 6x + C'(y) \Rightarrow C'(y) = 3y^2 \Rightarrow C(y) = y^3 + K \text{ (constant)}.$$

Hence,

$$\boxed{f(x, y) = x^3 + 6xy + y^3 + K \text{ (constant)}}$$

(ii) $F(x, y, z) = (e^x \sin y \cos z, e^x \cos y \cos z, -e^x \sin y \sin z).$

We seek a differentiable function $f : \mathbb{R}^3 \rightarrow \mathbb{R}$ such that $\nabla f(x, y, z) = \vec{F}(x, y, z)$ that is,

$$\begin{cases} f_x(x, y, z) = e^x \sin y \cos z \\ f_y(x, y, z) = e^x \cos y \cos z \\ f_z(x, y, z) = -e^x \sin y \sin z \end{cases}$$

Integrating the first equation w.r.t. x , we get

$$f(x, y, z) = \int e^x \sin y \cos z dx = e^x \sin y \cos z + C(y, z).$$

Now differentiating w.r.t. y , we get

$$e^x \cos y \cos z = f_y(x, y, z) = e^x \cos y \cos z + C_y(y, z) \Rightarrow C_y(y, z) = 0 \Rightarrow C(y, z) = K(z).$$

Hence,

$$f(x, y, z) = e^x \sin y \cos z + K(z) \quad (*)$$

We differentiate the new expression of f in $(*)$ w.r.t. z and get

$$-e^x \sin y \sin z = f_z(x, y, z) = -e^x \sin y \sin z + K'(z) \Rightarrow K'(z) = 0 \Rightarrow K(z) = K \text{ (constant)}.$$

Therefore, we find that

$$\boxed{f(x, y, z) = e^x \sin y \cos z + K \text{ (constant)}}$$

2. We recall that if $f : D \subset \mathbb{R}^2 \rightarrow \mathbb{R}$ is a continuous function and C is a regular curve contained in D , described by a vector function $\vec{r} : [a, b] \rightarrow \mathbb{R}^2$, then the line integral of the function f along the curve C w.r.t. the arclength parameter s is given by

$$\boxed{\int_C f(x, y) ds = \int_a^b f(\vec{r}(t)) \|\vec{r}'(t)\| dt}$$

and similarly for $f : D \subset \mathbb{R}^3 \rightarrow \mathbb{R}$.

Evaluate the line integrals:

- (a) $\int_C (x^2 + y^2) ds$ where C is the portion of the curve $x^2 + y^2 = 6y$ in the first quadrant joining $(0, 0)$ to $(3, 3)$.

Use the vector function $\vec{r}(t) := (3 \cos t, 3 + 3 \sin t)$ with $-\frac{\pi}{2} \leq t \leq 0$. So, $dx = -3 \sin t dt$, $dy = 3 \cos t dt$ and $ds = |\vec{r}'(t)| dt$.

$$\int_C (x^2 + y^2) ds = \int_C 6y ds = \int_{-\frac{\pi}{2}}^0 6(3 + 3 \sin t) \|\vec{r}'(t)\| dt = 54 \int_{-\frac{\pi}{2}}^0 (1 + \sin t) dt = 54 \left(\frac{\pi}{2} - 1 \right).$$

- (b) $\int_C y ds$ where C is the arc defined by $y = \sqrt{x}$, $0 \leq x \leq 2$;

The arc C is described by the vector function $\vec{r}(t) = (t, \sqrt{t})$ with $0 \leq t \leq 2$. Thus,

$$\begin{aligned} \int_C y ds &= \int_0^2 \sqrt{t} \|\vec{r}'(t)\| dt = \int_0^2 \sqrt{t} \sqrt{1 + \frac{1}{4t}} dt = \frac{1}{2} \int_0^2 \sqrt{4t+1} dt \\ &= \frac{1}{2} \left[\frac{1}{6} (4t+1)^{3/2} \right]_0^2 = \frac{26}{12}. \end{aligned}$$

- (c) $\int_C z^2 ds$ and $\int_C xy ds$, where is the curve described by the vector function $\vec{r} := [\frac{\pi}{2}, \frac{3\pi}{4}] \rightarrow \mathbb{R}^3$ defined by $\vec{r}(t) = (a \cos t, a \sin t, bt)$ for every $t \in [\frac{\pi}{2}, \frac{3\pi}{4}]$ with $(a, b) \neq (0, 0)$.

$$\vec{r}'(t) = (-a \sin t, a \cos t, b) \Rightarrow \|\vec{r}'(t)\| = \sqrt{a^2 + b^2}.$$

Thus

$$\begin{aligned} \int_C z^2 ds &= \int_{\frac{\pi}{2}}^{\frac{3\pi}{4}} b^2 t^2 \sqrt{a^2 + b^2} dt = b^2 \sqrt{a^2 + b^2} \int_{\frac{\pi}{2}}^{\frac{3\pi}{4}} t^2 dt \\ &= \frac{1}{3} b^2 \sqrt{a^2 + b^2} \left[\frac{27\pi^3}{64} - \frac{\pi^3}{8} \right] = \frac{19b^2\pi^3}{192} \sqrt{a^2 + b^2}. \end{aligned}$$

$$\begin{aligned} \int_C xy ds &= \int_{\frac{\pi}{2}}^{\frac{3\pi}{4}} a^2 \cos t \sin t \sqrt{a^2 + b^2} dt = a^2 \sqrt{a^2 + b^2} \int_{\frac{\pi}{2}}^{\frac{3\pi}{4}} \frac{1}{2} \sin(2t) dt \\ &= a^2 \sqrt{a^2 + b^2} \left[-\frac{1}{4} \cos(2t) \right]_{\frac{\pi}{2}}^{\frac{3\pi}{4}} = \frac{1}{4} a^2 \sqrt{a^2 + b^2} \left(-\cos\left(\frac{3\pi}{2}\right) + \cos(\pi) \right) \\ &= -\frac{1}{4} a^2 \sqrt{a^2 + b^2}. \end{aligned}$$

3. **(After the lecture on Friday)** Evaluate the line integral $\int_C \vec{F} \cdot d\vec{r}$ where

$\vec{F}(x, y) := \left(\frac{-y}{x^2 + y^2}, \frac{x}{x^2 + y^2} \right)$ and C is any circle centered at the origin and oriented clockwise.

Hint: First write down a parameterization of the curve C , that is, find a vector function describing the circle C with that orientation.

We can use the vector function $\vec{r}(t) := (R \cos t, -R \sin t)$ with $0 \leq t \leq 2\pi$ describing the circle C with a clockwise orientation. Notice that $x^2 + y^2 = R^2$. So,

$$\int_C \vec{F} \cdot d\vec{r} = \int_0^{2\pi} F(\vec{r}(t)) \cdot \vec{r}'(t) dt = \int_0^{2\pi} \frac{(R \sin t)(-R \sin t) + (R \cos t)(-R \cos t)}{R^2} dt = -2\pi.$$

4. Let C_1 be the upper half of the ellipse $\frac{(x-1)^2}{9} + \frac{y^2}{4} = 1$.

- (a) Sketch the curve C_1 .
- (b) $x = 1 + 3 \cos t$ and $y = 2 \sin t$ with $0 \leq t \leq \pi$.
- (c) Let C be the union of the curve C_1 and the x -axis. We call L the line segment on the x -axis joining $(-2, 0)$ to $(4, 0)$. So,

$$\int_C (x + 2y)dx + (2x + 3y)dy = \int_{C_1} (x + 2y)dx + (2x + 3y)dy + \int_L (x + 2y)dx + (2x + 3y)dy$$

$$\begin{aligned} & \int_{C_1} (x + 2y)dx + (2x + 3y)dy \\ &= \int_0^\pi [(1 + 3 \cos t + 4 \sin t)(-3 \sin t) + (2 + 6 \cos t + 6 \sin t)(2 \cos t)]dt = -6. \end{aligned}$$

$$\int_L (x + 2y)dx + (2x + 3y)dy = \int_{-2}^4 xdx = \left[\frac{1}{2}x^2 \right]_{-2}^4 = 6. \text{ Hence,}$$

$$\int_C (x + 2y)dx + (2x + 3y)dy = -6 + 6 = 0.$$

- (d) Notice that if D is the region enclosed by the curve C , from Green's Theorem

$$\int_C (x + 2y)dx + (2x + 3y)dy = \iint_D \left(\frac{\partial(2x + 3y)}{\partial x} - \frac{\partial(x + 2y)}{\partial y} \right) dxdy = 0.$$